PROJECT TITLE:
NATIONAL ACCOUNTS STATISTICS OF INDIA USING PYTHON

1. Project Overview

- The project titled **"National Accounts Statistics of India Using Python"** focuses on analysing India's National Accounts Statistics dataset using Python. The dataset was collected from the Government of India's Open Government Data Platform.
- The dataset contains economic information recorded at annual and quarterly frequencies, covering the period from 1999 to 2025–26. It includes important economic indicators such as GDP at current prices and constant prices, along with industry and institutional sector details.
- The project involves data understanding, data cleaning, preprocessing, feature engineering, descriptive statistical analysis, and exploratory data analysis (EDA). Various visualizations are created using Python libraries such as Matplotlib and Seaborn to understand economic patterns and trends.
- The main purpose of this project is to improve data quality and explore changes in India's economic indicators through statistical methods and visual representation.

2. Objectives of the Study

1. To understand the structure and characteristics of the National Accounts Statistics dataset of India, including its columns, data types, and economic indicators.
2. To clean and preprocess the dataset by identifying and handling missing values, removing duplicate records, correcting inconsistencies, standardizing column names, removing unnecessary columns, and converting data types where necessary.
3. To perform descriptive statistical analysis by calculating measures such as mean, median, and standard deviation for suitable numerical variables and examining economic indicators.
4. To analyse year-wise and quarterly changes in economic indicators, compare economic performance across available industries or sectors, and identify patterns and trends using visualizations created with Matplotlib and Seaborn.
5. To interpret the findings, summarize the major insights, and save the cleaned dataset in CSV format for further analysis and research.

3. Problem Statement

- National Accounts Statistics contains a large amount of economic data related to India's GDP, industries, and institutional sectors. However, analysing raw data directly can be difficult because it may contain missing values, duplicate records, inconsistent formats, and different data types.
- Without proper preprocessing and statistical analysis, identifying meaningful economic patterns and trends becomes challenging.
- Therefore, this project focuses on cleaning and organizing the National Accounts Statistics dataset using Python and applying statistical techniques and data visualization methods to understand changes in India's economic indicators over time.

4. Business Understanding

- National Accounts Statistics provides important information about the economic performance of India. GDP at current prices and constant prices are useful indicators for understanding the country's economic growth.
- This project aims to analyse these economic indicators using Python to identify trends, compare values across financial years, and understand the relationship between GDP at current prices and constant prices.
- The analysis helps present complex economic information in a simple and understandable form through statistical summaries and visualizations.

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5. Dataset Information

The dataset used in this project is titled **National Accounts Statistics of India**. It was collected from the Government of India's Open Government Data Platform.

Description	Details
Dataset Name	National Accounts Statistics of India
Source	Open Government Data Platform, Government of India
Location	India
Domain	National Accounts Statistics
Time Period	1999 to 2025–26
Data Frequency	Annual and Quarterly
Original Rows	7,836
Original Columns	13
Final Rows	7,821
Final Columns	16
File Format	CSV

The original dataset contains 7,836 records and 13 columns. After data cleaning and feature engineering, the final dataset contains 7,821 records and 16 columns.

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6. Data Dictionary

A data dictionary provides a description of each column in the dataset. It helps to understand the information stored in each variable.

S.No	Column Name	Description
1	Base_Year	Base year used for calculating economic values.
2	Series	Indicates the statistical series.
3	Year	Represents the financial or calendar year.
4	Indicator	Specifies the economic indicator.
5	Frequency	Indicates whether the data is annual or quarterly.
6	Revision	Represents the revision status of the data.
7	Industry	Identifies the industry category.
8	SubIndustry	Represents the sub-industry category.
9	Institutional_Sector	Identifies the institutional sector.
10	Quarter	Represents the quarter of the year.

11	Current_Price	Economic values measured at current prices.
12	Constant_Price	Economic values measured at constant prices.
13	Unit	Specifies the measurement unit.
14	Year_start	Extracted starting year from the year field.
15	Year_end	Extracted ending year from the year field.
16	Period_type	Identifies the type of period represented in the year field.

The last three columns were created during the feature engineering process.

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7. Tools & Technologies

Programming Language

- **Python:** Used for data cleaning, preprocessing, statistical analysis and data visualization.

Libraries

- **Pandas:** Used for data manipulation, handling missing values, removing duplicates and analysing tabular data.
- **NumPy:** Used for numerical calculations and statistical operations.
- **Matplotlib:** Used to create charts and graphs for visualizing economic trends.
- **Seaborn:** Used for statistical data visualization, including histograms, box plots, scatter plots and pair plots.

Development Environment

- **Google Colab:** Used as the cloud-based environment for writing and executing Python code.
- **CSV:** Used as the file format for storing and handling the dataset.

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8. Project Workflow

The project follows a systematic workflow to transform raw economic data into meaningful information.

Data Collection
↓
Initial Data Exploration
↓
Data Cleaning
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Data Validation
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Feature Engineering
↓
Final Data Exploration
↓
Descriptive Statistical Analysis
↓
Data Visualization
↓
Conclusion

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9. Data Pre-processing

Data preprocessing is an important step in data analysis. It helps improve data quality and prepares the dataset for statistical analysis and visualization.
The National Accounts Statistics dataset was inspected and cleaned using Python libraries, mainly Pandas and NumPy.

Initial Data Inspection

The following checks were performed to understand the dataset:

- Dataset shape
- First and last records
- Sample records
- Column names
- Data types
- Missing values
- Duplicate records
- Unique values
- Descriptive statistics

Dataset Shape

The original dataset contained:

- Rows: 7,836
- Columns: 13

Missing Value Handling

Missing values were identified using Pandas functions.
The missing values were handled to improve data quality and prepare the dataset for further analysis.

Duplicate Record Removal

Duplicate records were identified using the duplicated() function.
A total of 15 duplicate records were removed using the drop_duplicates() function.
After removing duplicates, the dataset contained 7,821 records.

Column Name Standardization

Column names were standardized to maintain a consistent naming format and make them easier to use during Python analysis.

Data Type Checking

The data types of the columns were checked to ensure that numerical and categorical variables were suitable for analysis.

Final Dataset

After data cleaning, the dataset contained:

- Rows: 7,821
- Columns: 13

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10. Feature Engineering

Feature engineering is the process of creating new columns from existing data to make analysis easier and more meaningful.
In this project, three new columns were created from the Year column.

New Column	Purpose
Year_start	Extracts the starting year from the financial year.

Year_end	Extracts the ending year from the financial year.
Period_type	Identifies the type of period represented in the year field.

These additional columns help organize the financial-year information and support year-wise analysis.

Final Dataset After Feature Engineering

Description	Count
Rows	7,821
Original Columns	13
New Columns	3
Final Columns	16

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11. Data Validation

Data validation was performed to check the quality and consistency of the cleaned dataset before proceeding with statistical analysis.

The following checks were considered:

- **Missing Value Check:** Checked whether any missing values remained after data cleaning.
- **Duplicate Check:** Verified duplicate records after removing 15 duplicate entries.
- **Data Type Check:** Examined column data types to ensure they were suitable for analysis.
- **Data Consistency:** Checked the consistency of values and formats in the dataset.

After preprocessing, the cleaned dataset was used for further statistical analysis and visualization.

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12. Statistical Analysis

Descriptive statistical analysis was performed to understand the numerical characteristics of the National Accounts Statistics dataset.

The analysis focused on measures of central tendency and variability, including Mean, Median and Standard Deviation.

Measures of Central Tendency

Mean and Median were calculated for GDP values at current prices and constant prices.

Measure	Current Price	Constant Price
Mean	2,197,954.19	1,281,543.94
Median	609,067.50	323,158.57

Interpretation:

The mean values are higher than the median values for both variables. This indicates that some high-value observations influence the average values.

Current-price GDP also has a higher mean and median than constant-price GDP.

Standard Deviation

Standard deviation was calculated to understand the variation in GDP values.

Variable	Standard Deviation

Current Price	4,584,887.05
Constant Price	2,908,893.67

Interpretation:
Current-price values have a higher standard deviation than constant-price values. This indicates greater absolute variation in current-price observations.

Indicator-wise Statistical Analysis
The dataset was grouped based on economic indicators. Mean, Median and Standard Deviation were calculated to understand the numerical characteristics of different indicators.
Interpretation:
Different economic indicators show different value ranges and statistical characteristics. This analysis helps compare their numerical distributions.

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13. Exploratory Data Analysis (EDA)
Exploratory Data Analysis (EDA) is used to understand the structure, characteristics and patterns in the dataset before and after data preprocessing.

Initial EDA
Initial EDA was performed on the original dataset to understand its structure and identify data quality issues.
The following activities were carried out:

- Checked the dataset shape.
- Viewed the first and last records.
- Examined sample records.
- Checked column names and data types.
- Identified missing values.
- Checked duplicate records.
- Examined unique values and numerical columns.
- Generated initial descriptive statistics.

The original dataset contained 7,836 rows and 13 columns. This initial exploration helped identify the preprocessing requirements.

Final EDA
Final EDA was performed after data cleaning and feature engineering to understand the prepared dataset.
The following activities were carried out:

- Examined the cleaned dataset structure.
- Verified missing values and duplicate records.
- Checked the newly created features.
- Explored numerical variables and economic indicators.
- Examined GDP values at current prices and constant prices.
- Studied year-wise patterns and relationships.
- Prepared visualizations to understand economic trends.

After preprocessing and feature engineering, the final dataset contained 7,821 rows and 16 columns.
The final EDA helped prepare the dataset for statistical analysis and visualization.

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14. Data Analysis and Visualization

Data analysis and visualization were performed to understand the characteristics, distributions, patterns and relationships in India's National Accounts Statistics dataset. The analysis focused on GDP at current prices and constant prices across 15 financial years, from 2011–12 to 2025–26.

The visual analysis was carried out using Matplotlib and Seaborn.

Univariate Analysis

Univariate analysis examines one variable at a time to understand its distribution and characteristics.

Histogram – Distribution of GDP at Current Prices

A histogram was created to understand the frequency distribution of GDP values at current prices.

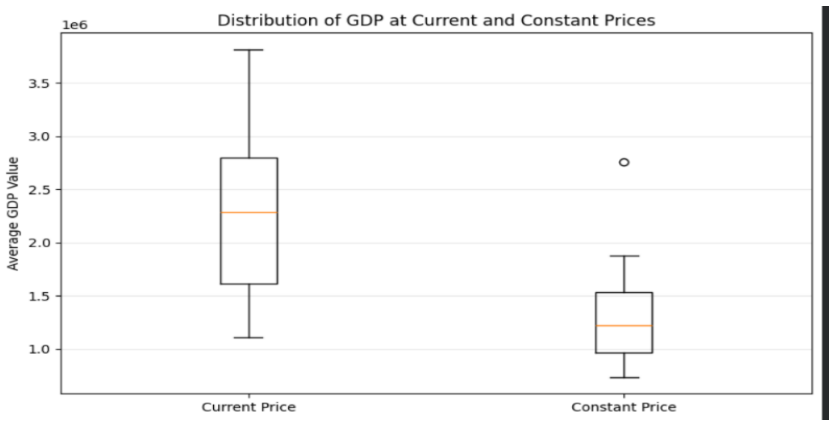


Finding:

The histogram shows how the 15 annual GDP observations are distributed across different value intervals. The values are spread across a range, with some observations having higher values than others.

Box Plot – GDP at Current and Constant Prices

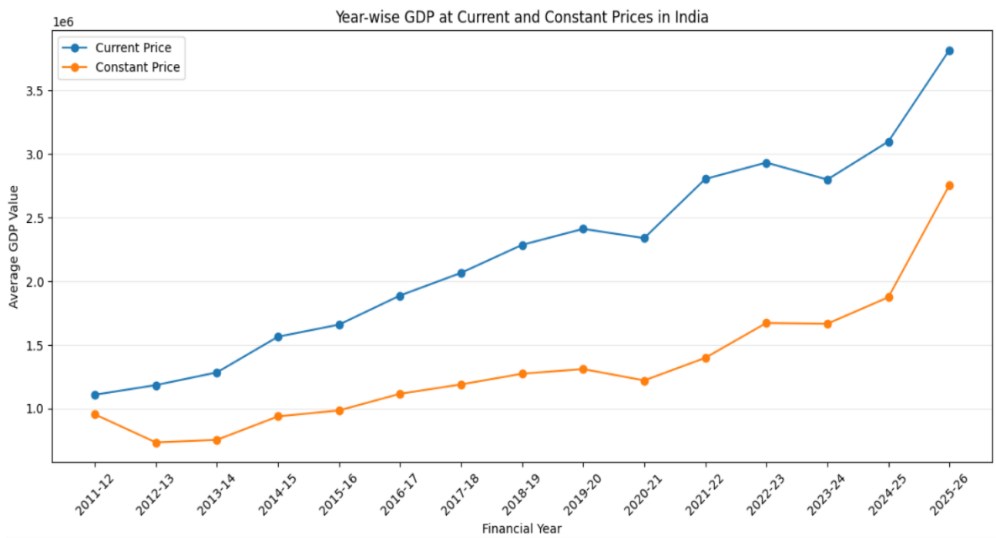
A box plot was created to compare the distribution, median, spread and possible outliers of current-price and constant-price GDP.



Finding:
Current-price GDP has a higher median and greater spread than constant-price GDP. The box plot also helps identify possible extreme observations.

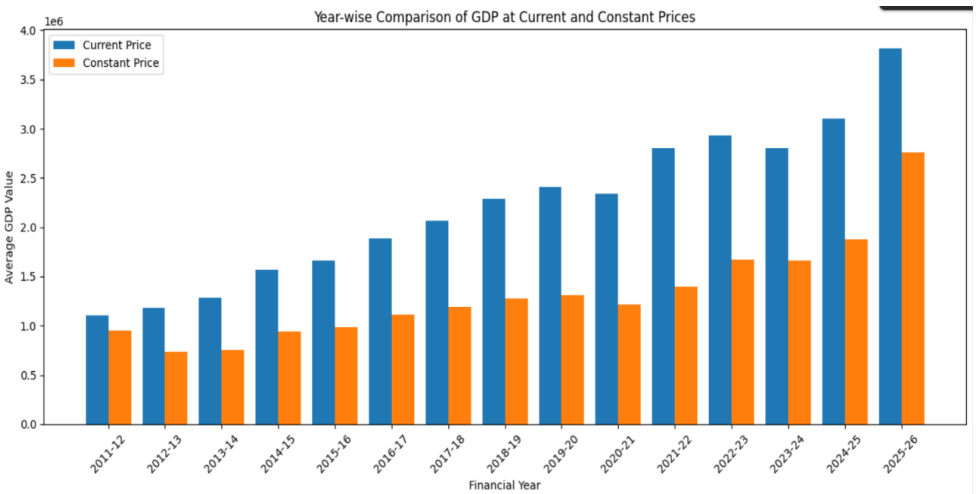
Bivariate Analysis
Bivariate analysis examines the relationship between two variables.

Line Chart – Year-wise GDP Trend
A line chart was created to analyse changes in GDP at current prices and constant prices across 15 financial years.



Finding:
Both GDP measures show an overall increasing trend from 2011–12 to 2025–26. However, decreases were observed during 2020–21 and 2023–24 compared with the respective previous years.

Grouped Bar Chart – Comparison of GDP
A grouped bar chart was used to compare current-price GDP and constant-price GDP for each financial year.

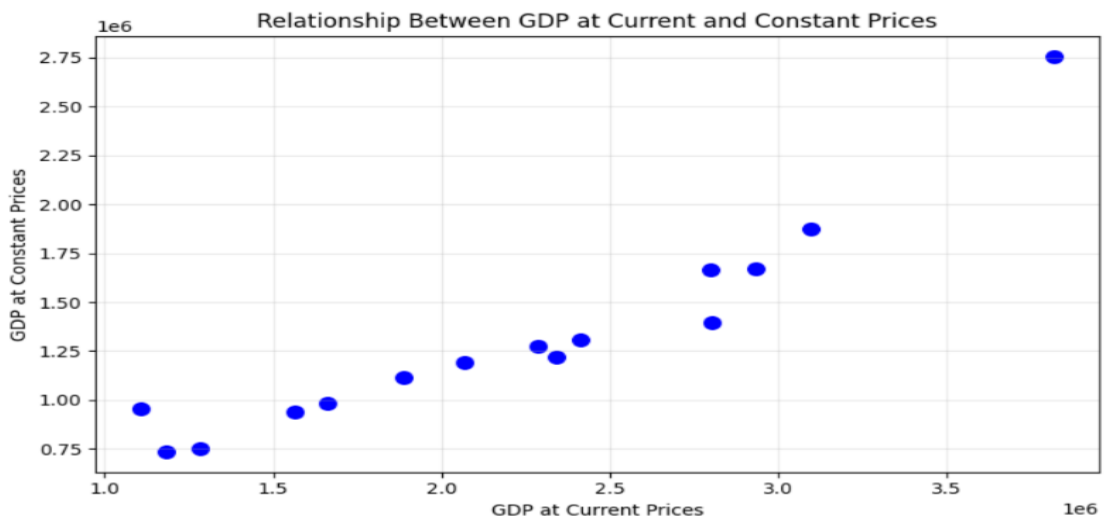


Finding:

Current-price GDP was higher than constant-price GDP throughout the selected 15 financial years. The chart makes the yearly differences between the two measures easier to understand.

Scatter Plot – Relationship Between GDP Measures

A scatter plot was created to examine the relationship between current-price GDP and constant-price GDP.



Finding:

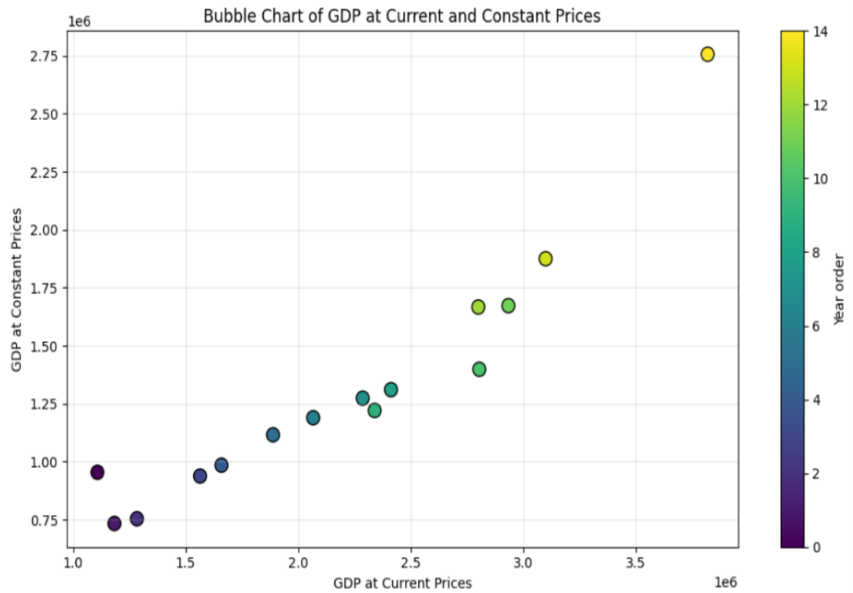
The points show an upward pattern, indicating a positive relationship between the two variables. The Pearson correlation coefficient was 0.9379, showing a strong positive linear relationship.

Multivariate Analysis

Multivariate analysis examines multiple variables together to understand their combined patterns and relationships.

Bubble Chart

A Bubble chart was created using current-price GDP and constant-price GDP, with colours representing the financial-year order.

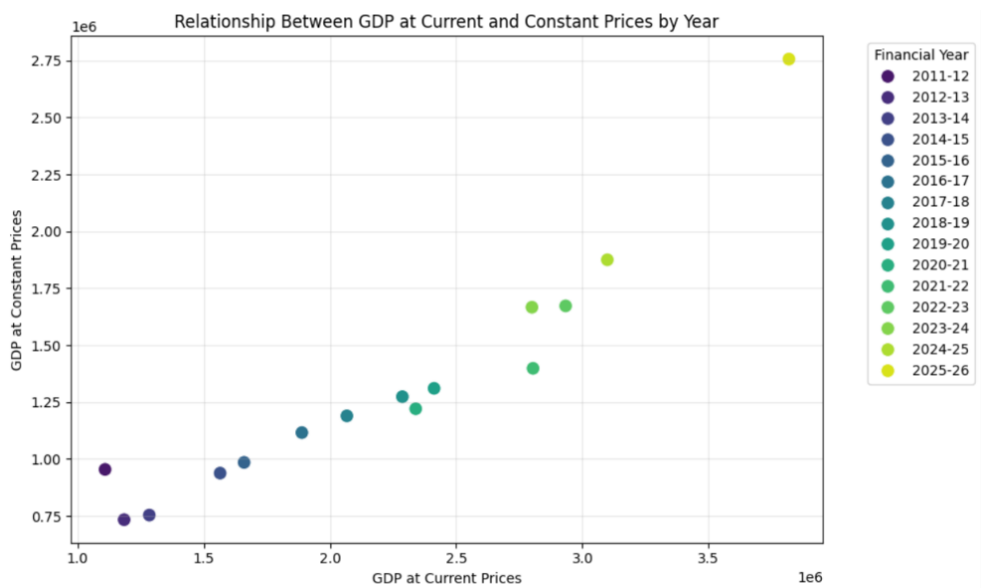


Finding:

The colour variation helps distinguish earlier and later financial years. Later years generally appear at higher GDP values.

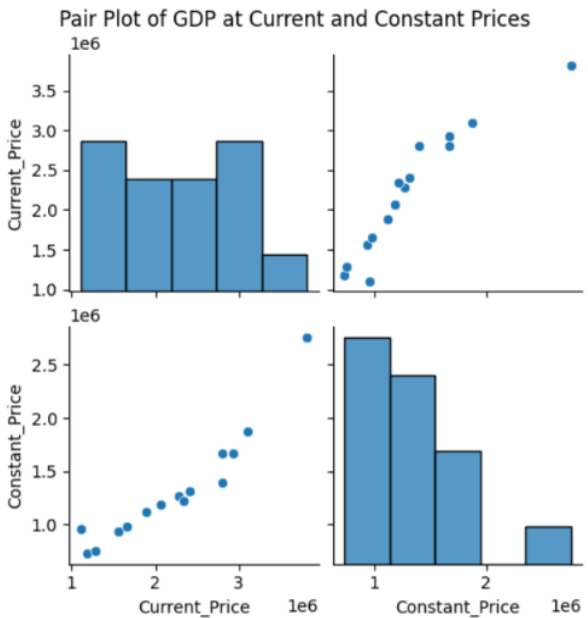
Multivariate Scatter Plot

A multivariate scatter plot was used to examine financial year, current-price GDP and constant-price GDP together.



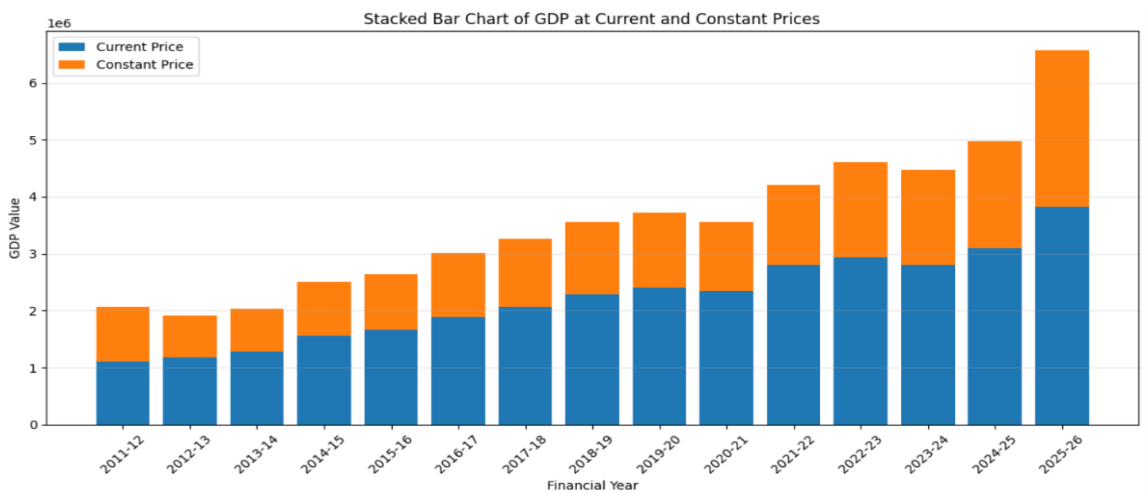
Finding:
The visualization helps observe the relationship between both GDP measures and their progression across financial years.

Pair Plot
A pair plot was created using Seaborn to examine the distributions and relationships between current-price GDP and constant-price GDP.



Finding:
The diagonal plots show the distribution of individual variables, while the other sections show their relationships through scatter plots. The visualization provides a combined view of distribution and relationship.

Stacked Bar Chart – Year-wise GDP Values
A stacked bar chart was created to display current-price GDP and constant-price GDP across financial years.



Finding:
The chart provides a visual display of both GDP measures across the selected years. The overall bar heights generally increase over time, with decreases visible during 2020–21 and 2023–24.
Note: The stacked height is only a visual combination of two alternative GDP measures and should not be interpreted as total GDP.

15. Key Findings

The analysis of National Accounts Statistics revealed the following important findings:

1. **Data Cleaning:** A total of 15 duplicate records were removed, reducing the dataset from 7,836 to 7,821 records.
2. **Dataset Structure:** After cleaning and feature engineering, the final dataset contained 7,821 rows and 16 columns.
3. **Descriptive Statistics:** The mean values were higher than the median values for both current-price and constant-price variables, indicating the influence of high-value observations.
4. **Variation:** Current-price GDP showed a higher standard deviation than constant-price GDP, indicating greater absolute variation in the observations.
5. **GDP Trend:** Both current-price and constant-price GDP generally increased across the 15 selected financial years.
6. **GDP Comparison:** Current-price GDP was higher than constant-price GDP throughout the selected period.
7. **Highest GDP Values:** Both measures reached their highest values in 2025–26. Current-price GDP was 3,818,356.32, while constant-price GDP was 2,756,158.21.
8. **Lowest GDP Values:** Current-price GDP recorded its lowest value in 2011–12 at 1,108,641.10, while constant-price GDP recorded its lowest value in 2012–13 at 733,691.06.
9. **Correlation Analysis:** The Pearson correlation coefficient was 0.9379, indicating a strong positive relationship between current-price and constant-price GDP.

16. Business Insights

The analysis provides useful insights into India's economic trends.

- The overall increase in GDP values indicates an upward trend across the selected financial years.
- The difference between current-price and constant-price GDP highlights the importance of analysing both measures.
- The higher standard deviation of current-price GDP indicates greater variation in its observations.
- The strong positive correlation suggests that both GDP measures generally move in the same direction.
- The decreases observed in 2020–21 and 2023–24 highlight the importance of examining year-wise economic changes rather than considering only the overall trend.

These insights demonstrate how statistical analysis and visualization can help understand economic data.

17. Recommendations and Suggestions

Based on the findings of this project, the following recommendations are suggested:

Regular Economic Data Analysis

Economic indicators should be analysed regularly to understand changes and identify important trends.

Compare Different GDP Measures

Current-price and constant-price GDP should be examined together to obtain a better understanding of economic performance.

Monitor Year-wise Changes

Years showing decreases in GDP values should be examined further to understand the possible reasons behind the changes.

Use Statistical Techniques

Statistical measures such as mean, median, standard deviation and correlation should be used to understand economic data more effectively.

Improve Future Analysis

Future studies can include additional economic indicators, industry-wise comparisons and forecasting techniques to explore economic trends in greater detail.

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18. Project Outcome

The project successfully demonstrates how Python can be used to analyse National Accounts Statistics of India and transform raw economic data into meaningful information.

The major outcomes of the project include:

- Successfully cleaned and preprocessed the economic dataset.
- Removed duplicate records and handled missing values.
- Created three new features using feature engineering.
- Performed descriptive statistical analysis using mean, median and standard deviation.
- Analysed GDP at current prices and constant prices across 15 financial years.
- Created different visualizations to understand economic patterns and relationships.
- Identified an overall increasing trend in GDP values.
- Found a strong positive correlation of 0.9379 between current-price GDP and constant-price GDP.
- Prepared a cleaned dataset for further analysis.

The project improved the understanding of practical data analysis using Python and statistical techniques.

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19. Conclusion

This project demonstrates the use of Python for analysing India's National Accounts Statistics.

The analysis involved data collection, data cleaning, preprocessing, feature engineering, descriptive statistical analysis and data visualization.

The findings showed that GDP at current prices and constant prices generally increased across the selected financial years. The correlation analysis also revealed a strong positive relationship between the two GDP measures.

Overall, this project helped transform raw economic data into meaningful information through statistical analysis and visualization.